MILITARY STANDARD

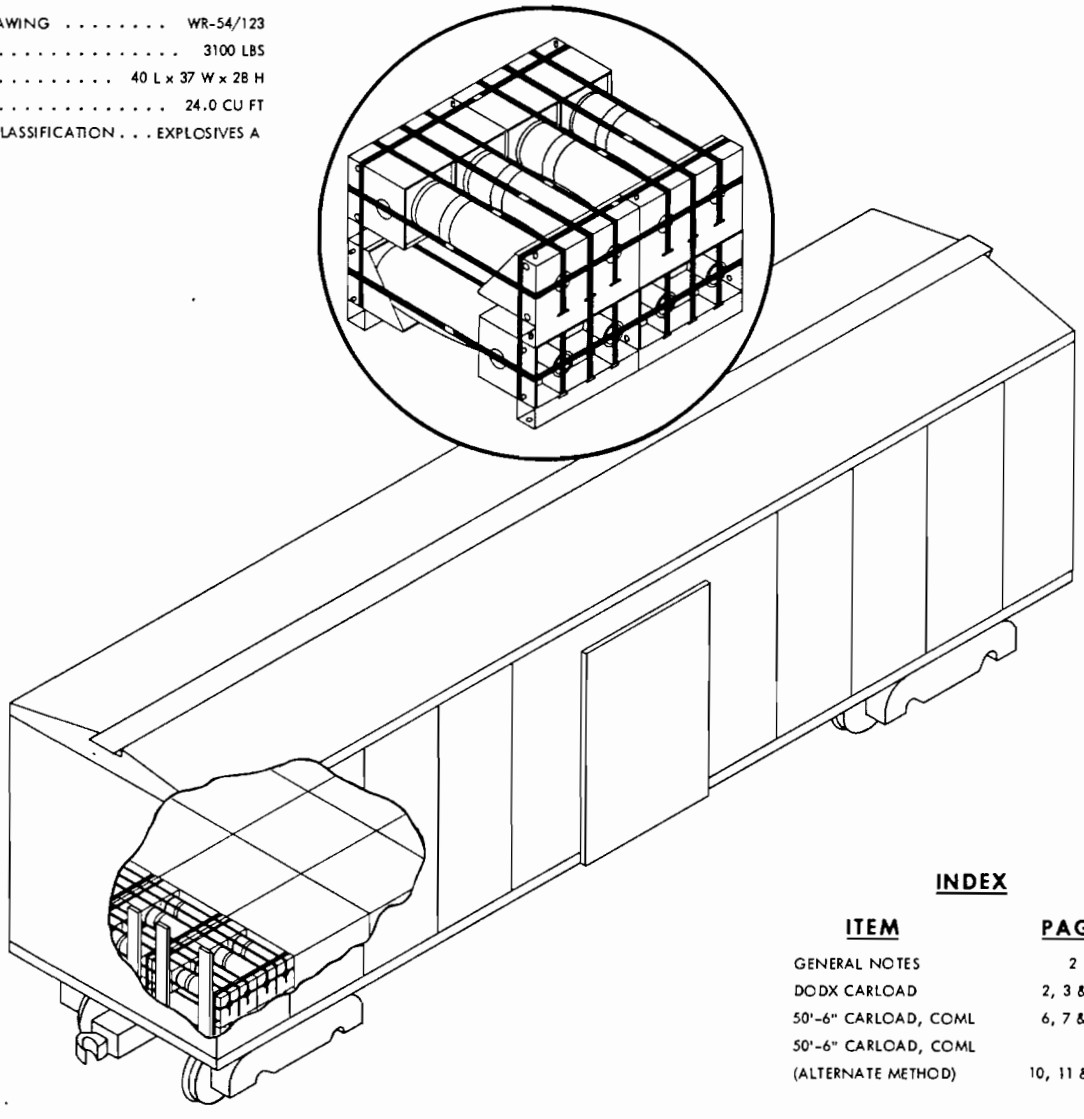
CARLOADING
PROJECTILE, 8 INCH MK 21 & MODS
IN ADAPTER MK 74 MOD 0
FLEET ISSUE UNIT LOAD

MIL-STD-1325-81 (NAVY)

17 JULY 1973
SUPERSEDING
WR-52/B1
12 OCTOBER, 1967

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/123
WEIGHT . . . . . 3100 LBS
DIMENSIONS . . . . . 40 L x 37 W x 28 H
CUBE . . . . . 24.0 CU FT
DOT HAZARD CLASSIFICATION . . . EXPLOSIVES A



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2, 3 & 4
50'-6" CARLOAD, COML	6, 7 & 8
50'-6" CARLOAD, COML (ALTERNATE METHOD)	10, 11 & 12

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK 236 (NAVY)

AUTHORIZED AND RELEASED FOR GENERAL USE.

M. S. Gray NWHL 7/3/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 7/3/73
SIGNATURE (CRD) SYSCOM, BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck MILITARY ASSISTANT 3/23/73
SIGNATURE DATE

ORIGINATOR L. R. Caringi 7/2/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 112 INCHES OR GREATER INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 8 INCH PROJECTILES MK 21 & MODS UNITIZED IN ADAPTER MK 74 MOD 0 IN ACCORDANCE WITH WR-54/123.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:

   DUNNAGE LUMBER - FED.SPEC. MM-L-751

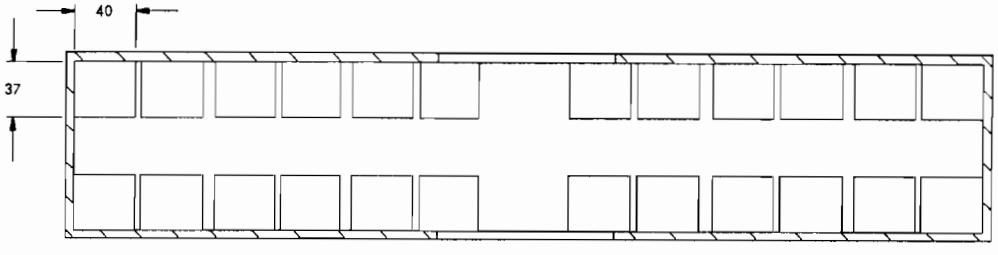
   NAILS - FED. SPEC. FF-N-105

   STRAPPING - FED. SPEC. QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)

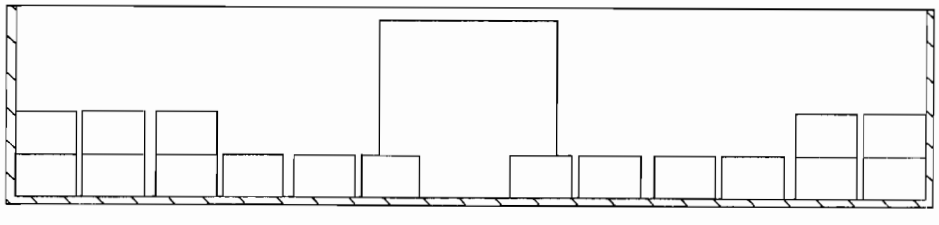
   SEALS - FED. SPEC. QQ-S-781, STYLE III, HEAVY DUTY
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL. I).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 34 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH SEE MIL-STD-1325 (NAVY). CROSSMEMBERS WILL BE USED IN SUCH A MANNER THAT THE TOTAL NUMBER OF CROSS-MEMBERS RETAINING ANY TWO STACKS WILL BE SUFFICIENT TO LIMIT THE AVERAGE LOAD RETAINED BY EACH TO NOT MORE THAN 2500 LBS.

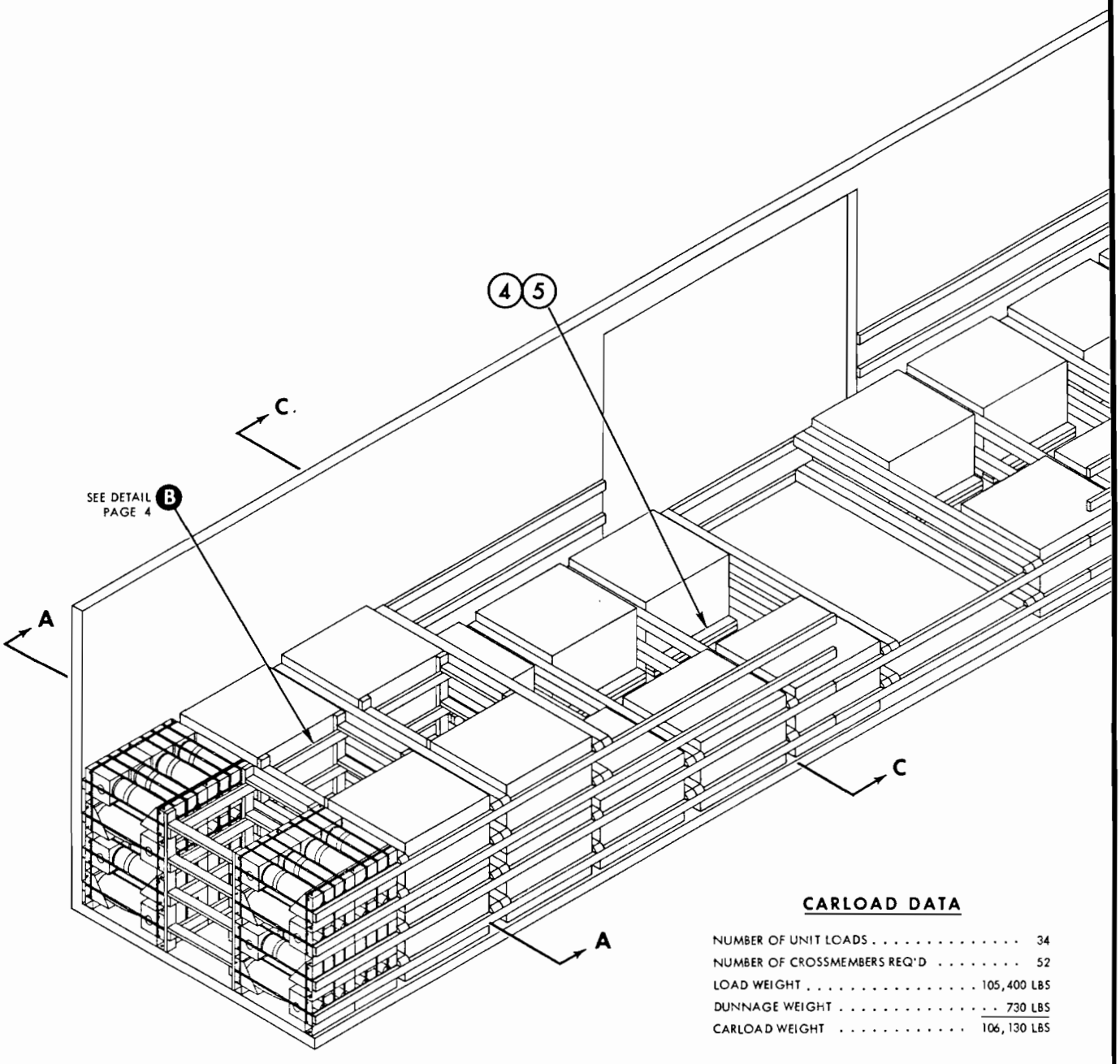


LOADING PLAN



LOADING PLAN
ELEVATION

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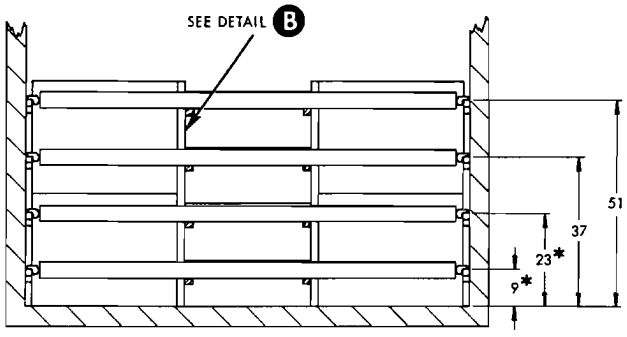
CARLOAD DATA

NUMBER OF UNIT LOADS	34
NUMBER OF CROSSMEMBERS REQ'D	52
LOAD WEIGHT	105,400 LBS
DUNNAGE WEIGHT	730 LBS
CARLOAD WEIGHT	106,130 LBS

LIST OF MATERIALS & NAILING DATA

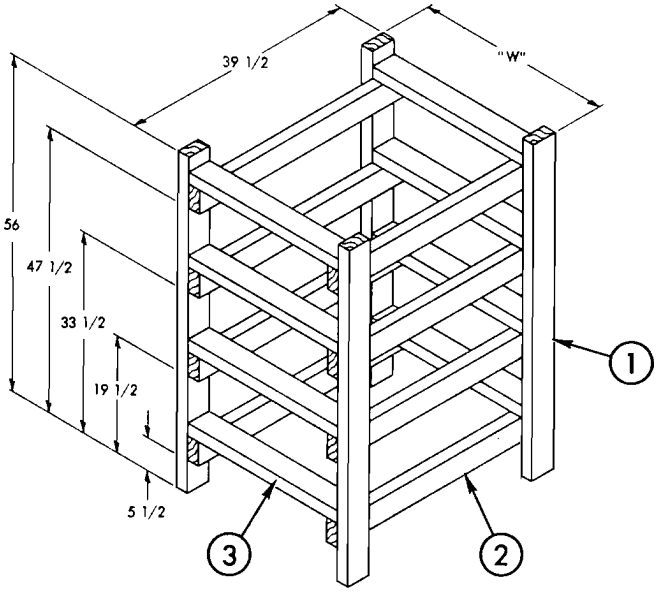
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
5	SLEEPER	2 x 4 x 40	14	4	2 PER FOOT	30d
4	SLEEPER	2 x 4 x 40	14	CAR FLOOR	2 PER FOOT	16d
3	CROSS BRACE	2 x 4 x TO SUIT	40	TOENAIL 2	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 x 4 x 39 1/2	40	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 x 4 x 56	20	SEE 2	-	-

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SECTION A-A

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSSMEMBERS
*DOORWAY MEMBER LOCATIONS

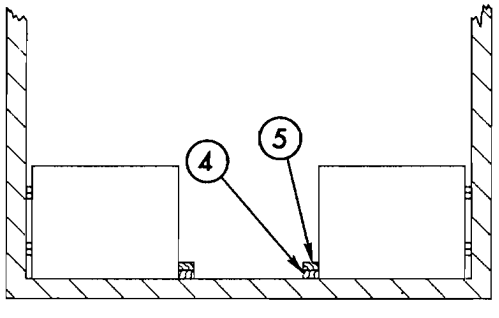


"W" = DISTANCE BETWEEN UNIT LOADS ACROSS THE CAR

SWAY BRACE ASSEMBLY DETAIL A PAGE 7 TOGETHER WITH SLEEPERS (PIECES 4 & 5) MAY BE USED IN PLACE OF DETAIL B (PIECES 1, 2 & 3)

DETAIL B

SWAY BRACE ASSEMBLY
5 REQUIRED



SECTION C-C

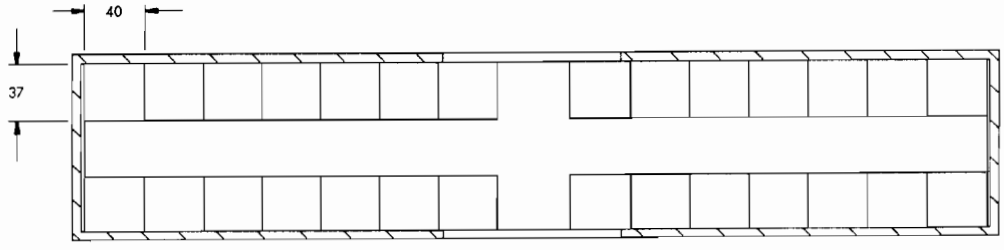
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 36 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 23 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 23 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL H) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 25 AND 26 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

POSITION BASE END ADAPTERS OF THE BOTTOM LAYER OF EACH UNIT LOAD AGAINST THE SIDEWALLS OF THE BOXCAR

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
26	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	4	25	2 PER JOINT	10d
25	VERTICAL DOORWAY MEMBER	2 x 3 x 48	4	DOOR POST	2 PER FOOT	20d
24	HOLD DOWN CLEAT	2 x 6 x 60	2	23	2 PER FOOT	10d
23	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
22	STRUT	2 x 6 x WEDGE FIT	24	19	3 PER JOINT	16d
21	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH-1	2	19	3 PER JOINT	10d
20	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	4	19	3 PER JOINT	10d
19	CENTER GATE VERTICAL	2 x 6 x 32	8	SEE 18	-	-
18	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	4	19	3 PER JOINT	10d
17	HORIZONTAL BACK-UP CLEAT	2 x 6 x 30	8	12 ,	14	16d
16	DIAGONAL BRACE	4 x 4 x 50 SEE DETAIL D	8	8, 12	1 EACH END	60d
15	CENTER CLEAT	2 x 4 x 36	4	8	7	16d
14	HORIZONTAL POCKET CLEAT	2 x 6 x 18	8	12	7	10d

LIST OF MATERIALS & NAILING DATA

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
13	VERTICAL BACK-UP CLEAT	2 x 6 x 27	4	CAR WALL	8	10d
12	HORIZONTAL WALL CLEAT	2 x 6 x 72	8	CAR WALL	16	10d
11	UPPER WALL CLEAT	2 x 4 x 18	4	CAR WALL	5	10d
10	INTERMEDIATE WALL CLEAT	2 x 4 x 10 3/8	4	CAR WALL	4	10d
9	LOWER WALL CLEAT	2 x 4 x 7 1/4	4	CAR WALL	4	10d
8	CROSS BRACE	4 x 4 x CAR WIDTH-1	4	SEE 7, 15, 16	-	-
7	CROSS BRACE STIFFENER	2 x 6 x CAR WIDTH-1	4	8	2 PER FOOT	10d
6	CROSS BRACE	2 x 4 x TO SUIT	8	4	2 PER FOOT	10d
5	LONGITUDINAL BRACE	2 x 4 x 40	8	4	3 PER JOINT	10d
4	SUPPORT	2 x 4 x (X+3 1/2)	8	SEE 5, 6	-	-
3	SLEEPER	2 x 4 x 40	28	2	2 PER FOOT	30d
2	SLEEPER	2 x 4 x 40	28	CAR FLOOR	2 PER FOOT	16d
1	END WALL MEMBER	2 x 6 x 60	8	CAR WALL	1 PER FOOT	10d

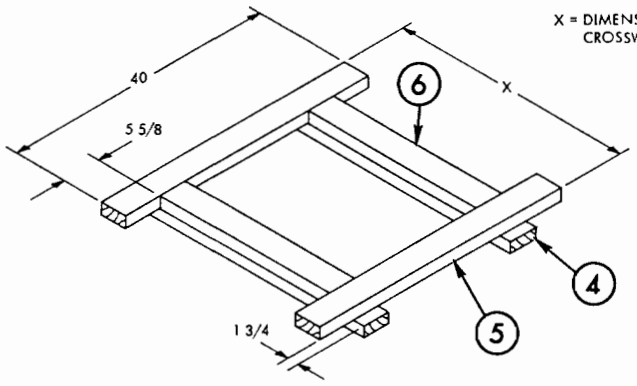
LIST OF MATERIALS & NAILING DATA

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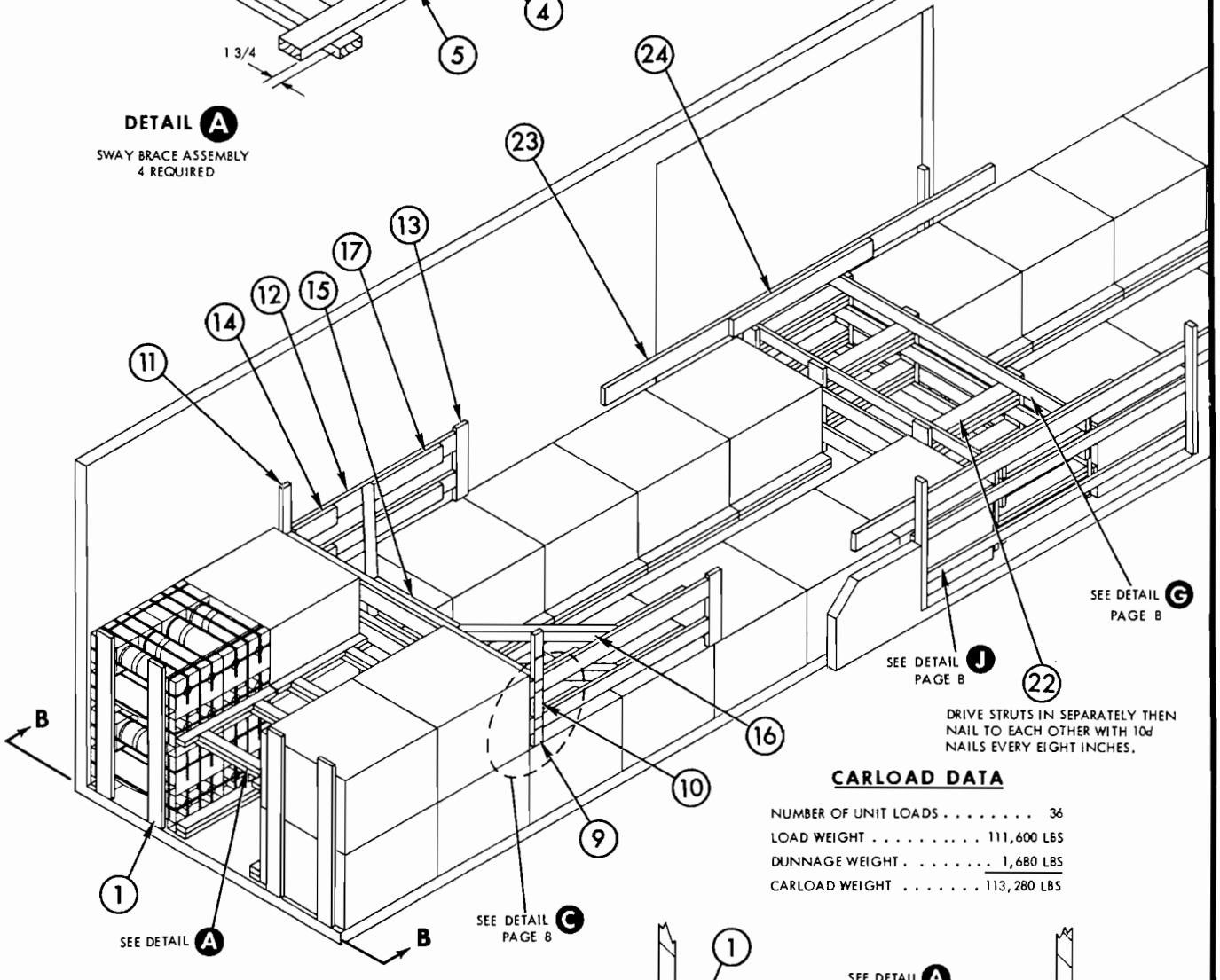
X = DIMENSION BETWEEN UNIT LOADS CROSSWISE IN CAR. (36 IN. REF)

DETAIL A MUST BE BUILT IN PLACE. INSERT PIECE 4 APPROXIMATELY 1 3/4" INTO THE POCKETS OF ADJACENT NOSE END ADAPTERS, AND NAIL PIECES 5 & 6 AS SHOWN.

SWAY BRACE ASSEMBLY DETAIL B, PAGE 4, MAY BE USED IN PLACE OF PIECES 2, 3, 4, 5, & 6



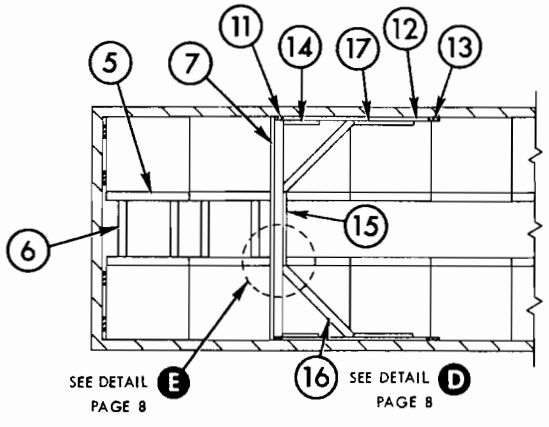
DETAIL A
SWAY BRACE ASSEMBLY
4 REQUIRED



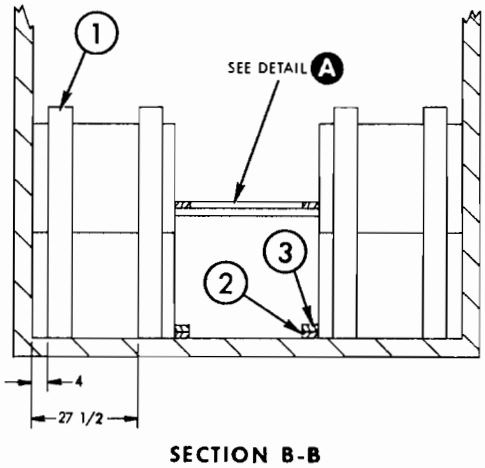
DRIVE STRUTS IN SEPARATELY THEN NAIL TO EACH OTHER WITH 10d NAILS EVERY EIGHT INCHES.

CARLOAD DATA

NUMBER OF UNIT LOADS	36
LOAD WEIGHT	111,600 LBS
DUNNAGE WEIGHT	1,680 LBS
CARLOAD WEIGHT	113,280 LBS

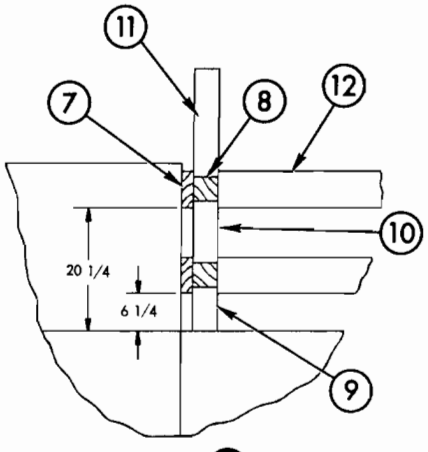


PARTIAL PLAN VIEW

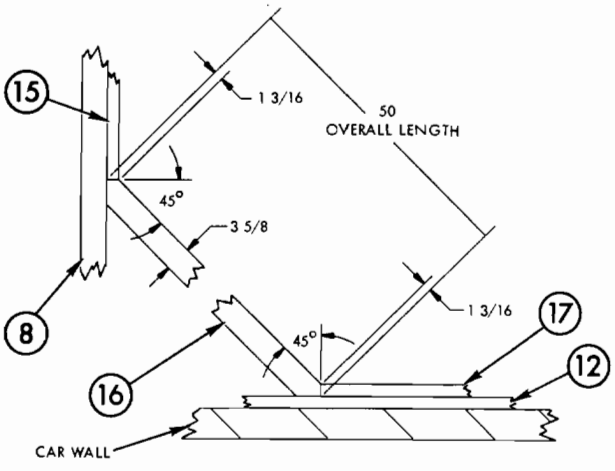


SECTION B-B
SHOWING LOCATION OF END WALL MEMBERS

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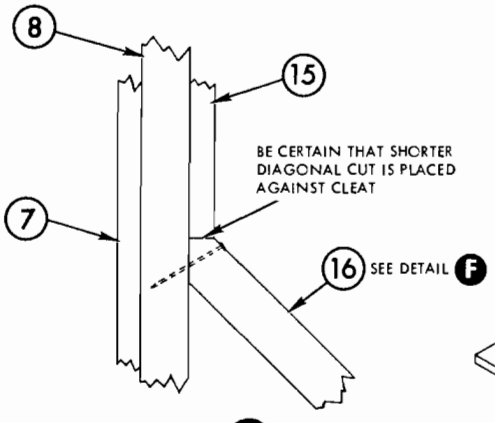


DETAIL C



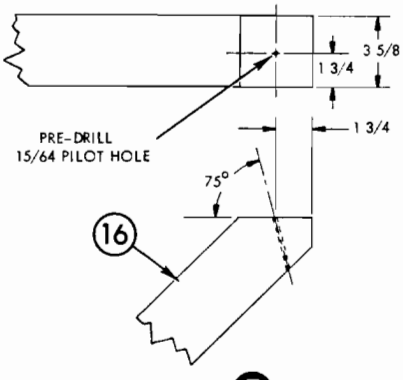
DETAIL D

POSITIONING OF DIAGONAL BRACE



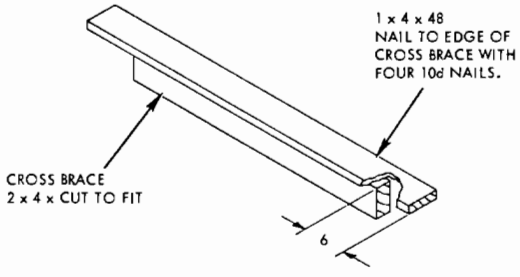
DETAIL E

POSITIONING OF DIAGONAL BRACE



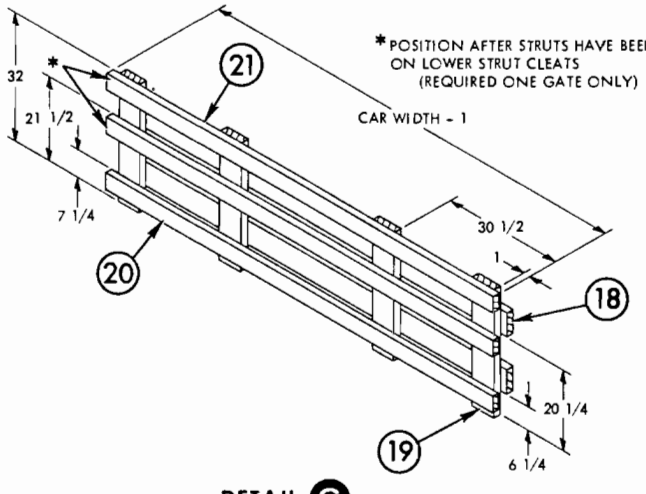
DETAIL F

LOCATION OF PILOT HOLE FOR 60d NAIL



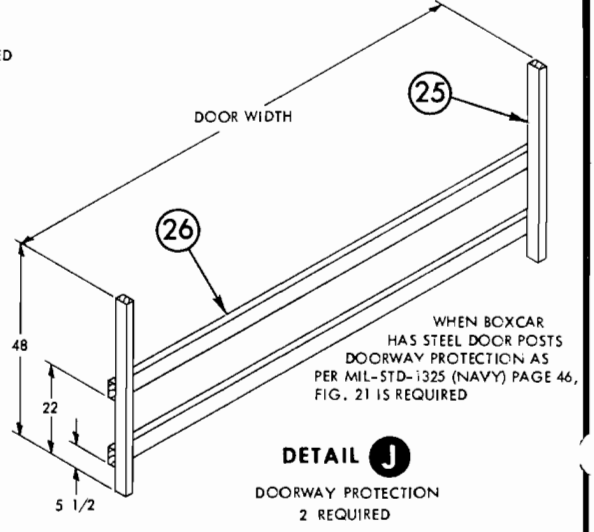
DETAIL H

TOP SWAY BRACE



DETAIL G

CENTER GATE



DETAIL J

DOORWAY PROTECTION
2 REQUIRED

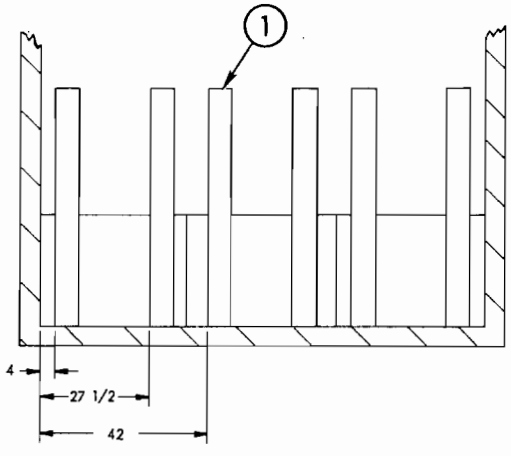
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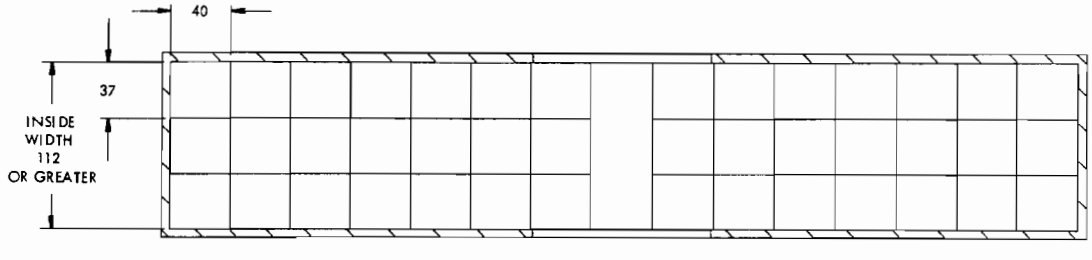
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50 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF 42 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. THIS ALTERNATE LOADING PLAN MAY ONLY BE USED WHEN A BOXCAR IS AVAILABLE WITH A 9 FT 4 IN OR GREATER INSIDE WIDTH AND A LOAD LIMIT IN EXCESS OF THE CARLOAD WEIGHT (131,030 LBS).
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 7 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 7 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS. DOORWAY PROTECTION, PIECES 9 AND 10 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



SECTION A-A
SHOWING LOCATION OF END WALL MEMBERS

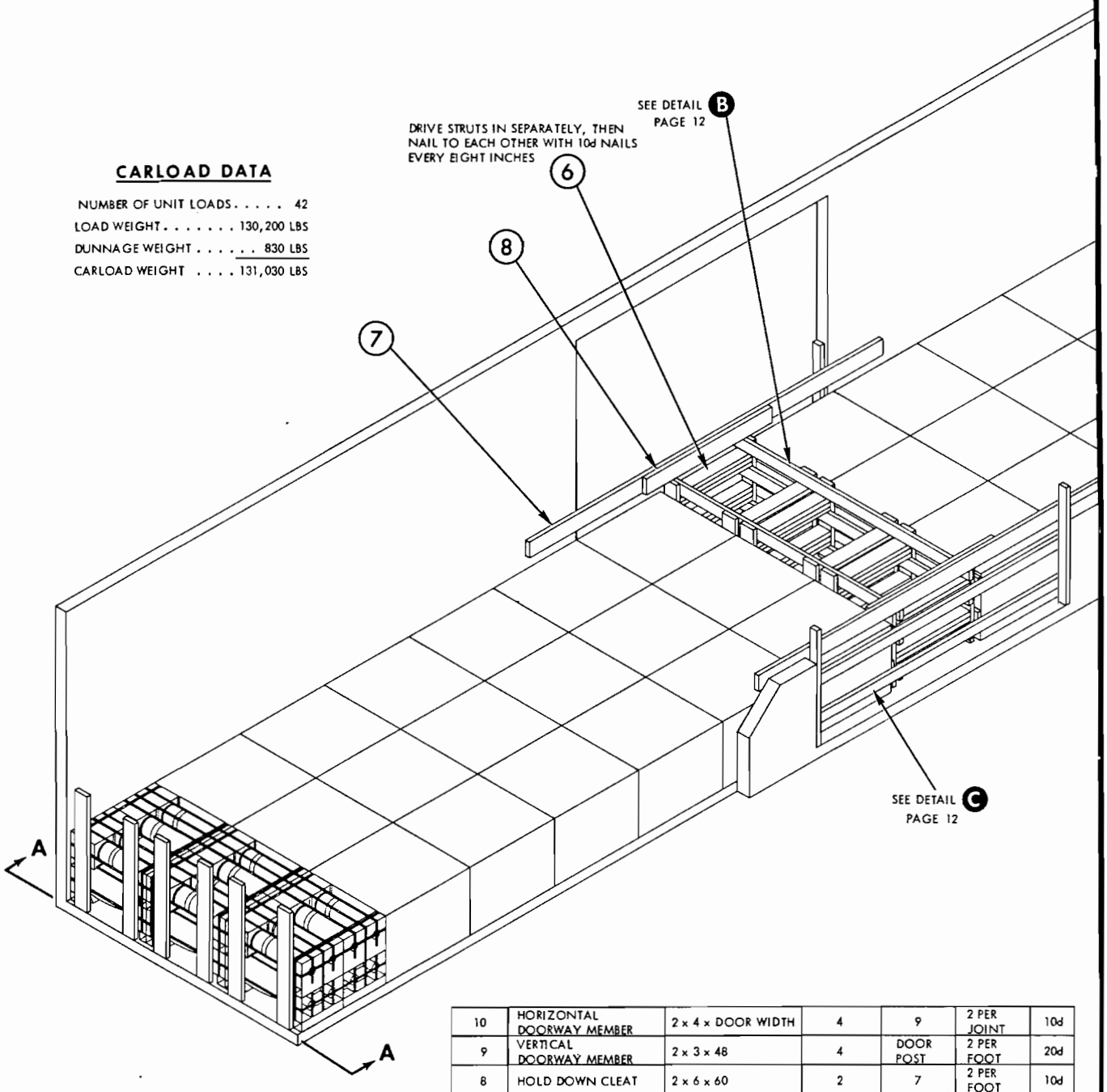


LOADING PLAN

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:01Z
Check the source to verify that this is the current version before use.

CARLOAD DATA

NUMBER OF UNIT LOADS 42
LOAD WEIGHT . . . . . 130,200 LBS
DUNNAGE WEIGHT . . . . . 830 LBS
CARLOAD WEIGHT . . . . 131,030 LBS

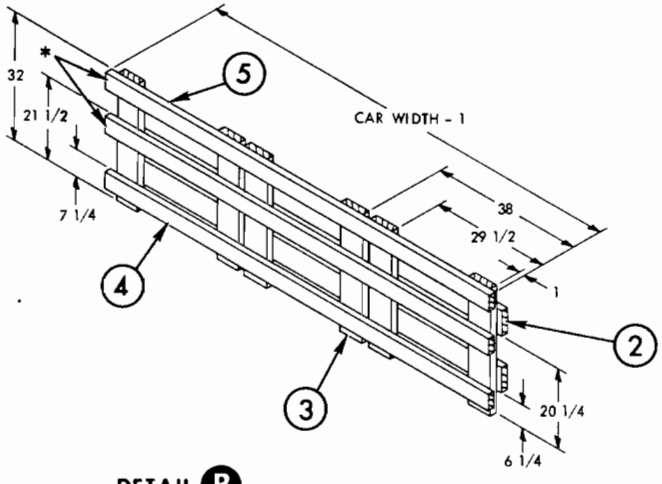


PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
10	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	4	9	2 PER JOINT	10d
9	VERTICAL DOORWAY MEMBER	2 x 3 x 48	4	DOOR POST	2 PER FOOT	20d
8	HOLD DOWN CLEAT	2 x 6 x 60	2	7	2 PER FOOT	10d
7	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
6	STRUT	2 x 6 x WEDGE FIT	36	3	3 PER JOINT	16d
5	CENTER GATE HORIZONTAL	2 x 4 x CAR WIDTH - 1	2	3	3 PER JOINT	10d
4	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH - 1	4	3	3 PER JOINT	10d
3	CENTER GATE VERTICAL	2 x 6 x 32	12	SEE 2	-	-
2	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH - 1	4	3	3 PER JOINT	10d
1	END WALL MEMBER	2 x 6 x 48	12	CAR WALL	1 PER FOOT	10d

LIST OF MATERIALS & NAILING DATA

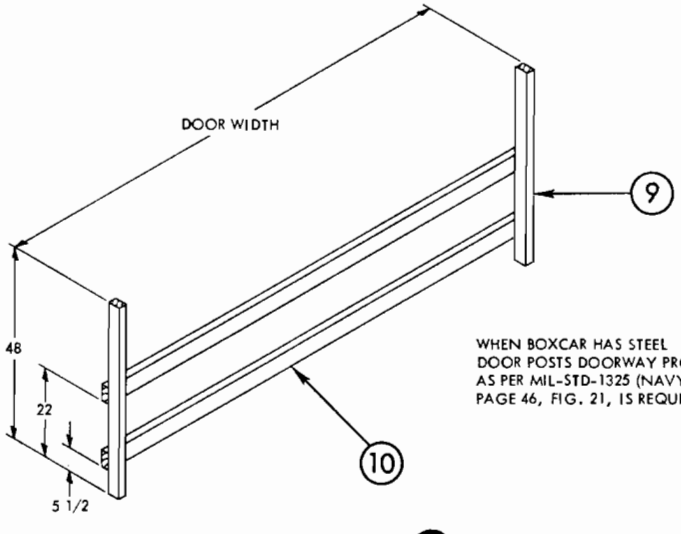
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* POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)



DETAIL B

CENTER GATE
2 REQUIRED



WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21, IS REQUIRED.

DETAIL C

DOORWAY PROTECTION
2 REQUIRED

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